



Management Science

Publication details, including instructions for authors and subscription information:
<http://pubsonline.informs.org>

Communication Quality and the Cost of Language: Evidence from Stack Overflow

Jacopo Bregolin

To cite this article:

Jacopo Bregolin (2025) Communication Quality and the Cost of Language: Evidence from Stack Overflow.
Management Science 71(6):4669-4687. <https://doi.org/10.1287/mnsc.2022.01333>

Full terms and conditions of use: <https://pubsonline.informs.org/Publications/Librarians-Portal/PubsOnLine-Terms-and-Conditions>

This article may be used only for the purposes of research, teaching, and/or private study. Commercial use or systematic downloading (by robots or other automatic processes) is prohibited without explicit Publisher approval, unless otherwise noted. For more information, contact permissions@informs.org.

The Publisher does not warrant or guarantee the article's accuracy, completeness, merchantability, fitness for a particular purpose, or non-infringement. Descriptions of, or references to, products or publications, or inclusion of an advertisement in this article, neither constitutes nor implies a guarantee, endorsement, or support of claims made of that product, publication, or service.

Copyright © 2024, INFORMS

Please scroll down for article—it is on subsequent pages



With 12,500 members from nearly 90 countries, INFORMS is the largest international association of operations research (O.R.) and analytics professionals and students. INFORMS provides unique networking and learning opportunities for individual professionals, and organizations of all types and sizes, to better understand and use O.R. and analytics tools and methods to transform strategic visions and achieve better outcomes. For more information on INFORMS, its publications, membership, or meetings visit <http://www.informs.org>

Communication Quality and the Cost of Language: Evidence from Stack Overflow

Jacopo Bregolin^a

^aUniversity of Liverpool, Liverpool L69 7ZH, United Kingdom

Contact: jacopo.bregolin@liverpool.ac.uk,  <https://orcid.org/0000-0003-2873-5191> (JB)

Received: May 3, 2022

Revised: November 10, 2023; February 19, 2024

Accepted: March 7, 2024

Published Online in Articles in Advance:
September 17, 2024

<https://doi.org/10.1287/mnsc.2022.01333>

Copyright: © 2024 INFORMS

Abstract. The transmission of information is crucial for productivity and growth. However, language differences may limit its effectiveness. This is particularly relevant for knowledge platforms that aim to be global, given the cultural diversity of the pool of users. In this paper, I empirically investigate how the exogenous cost of language affects communication quality and the trade-offs faced by knowledge platforms in implementing their website in multiple languages. I exploit the staggered introduction of websites for languages other than English on a question-and-answer platform to demonstrate two main facts. First, nonnative English speakers who contribute in English increase their answers' quality once able to use their native language, and their answers are more likely to solve the questioner's problem. The native-language answers drive the effect, which is larger when the question quality is higher and the incentives are stronger. Second, users who participate in their native language but not in English contribute lower-quality answers compared with those who contribute in English. This suggests that lower language barriers allow users with less expertise in the subject to participate. These results show that the platform should adopt multiple languages to maximise the quality of the information collected, although doing so may also result in an inflow of low-quality content from new users.

History: Accepted by Joshua Gans, business strategy.

Supplemental Material: The online appendix and data files are available at <https://doi.org/10.1287/mnsc.2022.01333>.

Keywords: cost of language • communication quality • knowledge platforms

1. Introduction

Communication as a way to transmit information is fundamental for the development of our society. The sharing of knowledge allows people to take advantage of each others' human capital investments, speeding up learning and productivity. Knowledge platforms that aggregate information, like Wikipedia and Stack Overflow, play a crucial role in this respect, as they can reach large audiences and leverage a global pool of knowledge. However, both online and offline, the quality of the shared information can be limited in several ways. On the one hand, communication can be affected by the too little incentives of the information holder. On the other hand, exogenous cognitive boundaries, such as the lack of proficiency in the language used, can constrain our ability to share information.¹ These issues are detrimental to knowledge platforms, as content quality determines their success.² Some platforms use incentive systems such as virtual rewards to address the first problem and allow their users to communicate in several languages to address the second problem. Whereas a large literature investigates the effectiveness of incentives, research on the identification of platforms' trade-offs when implementing their sites in multiple languages is sparser.

This paper contributes to the literature by investigating the quality trade-offs that knowledge platforms face when they allow their users to use multiple languages rather than just English. Intuitively, lower language barriers may let existing users provide higher-quality contributions and let new users join the platform. Quality trade-offs arise if new users have less expertise and provide lower-quality content. This paper tests these hypotheses, studying if and to what extent existing users provide higher-quality information when they are allowed to use their native language rather than English, and whether new users provide lower-quality contributions compared with existing users.

The analysis leverages data from Stack Overflow, an English-based question-and-answer (Q&A) website focused on topics related to computer programming. The platform introduced non-English versions of the website in a staggered fashion, allowing the researcher to observe contributions from two sets of users: those who were active in English and started to contribute in their native language once it became available, and those that registered in the English site but did not contribute, and only participated in their native language when allowed. I refer to the first set of users as *old*

joiners, and to the second set of users as *new joiners*. In particular, the researcher can observe old joiners' contributions before and after their native language became available, and compare old joiners' contributions with those of new joiners within the native-language sites.

This paper shows two main facts. First, once old joiners are allowed to use their native language, both their answers' quality and the probability that their answers solve the questioner's problem increase. These effects are driven by answers written in the native language, and increase with answerer incentives and the question quality. Second, new joiners' contributions in non-English languages are of lower quality compared with old joiners' contributions, possibly because of new joiners' lower expertise. These results suggest that by implementing multiple languages, the platform reduces users' communication costs. This encourages users to produce quality content but also allows for the participation of inexperienced contributors.

I illustrate the trade-off with a simple theoretical framework of communication between two users, Bob and Alice, where Alice may provide the information that Bob needs to complete a task. Alice decides her communication quality based on how much she internalises Bob's utility, the question's quality, her expertise, and the cost of language.³ The framework highlights two aspects. First, the quality of Alice's answer depends on her cost of using the language available, which is potentially high if it is not her native language. Second, her participation is subject to having sufficiently high expertise relative to the language cost. It follows that, for certain levels of expertise and cost, the availability of her native language would (1) increase her answer's quality if she has high expertise or (2) allow her to participate if she has low expertise. Overall, the framework suggests that introducing multiple languages would increase the quality of contributions from existing users. However, it would also allow for the participation of new contributors who may provide lower-quality content.

To test these predictions, I use the staggered introduction of new language sites by Stack Overflow. The platform was created in 2008 in English; over time, it sequentially implemented additional websites in Russian, Portuguese, Japanese, and Spanish with the same purpose and function as the original website. A unique identifier for each user allows the researcher to observe users' contributions across sites.

To proxy for communication quality, I look at two empirical measures. One measure is based on the users' communication content and text characteristics; the second one measures communication outcomes. More precisely, the former corresponds to the number of separate snippets of code included in the answer. Because questions relate to computer programming, a more developed and informative answer would include a

step-by-step procedure that alternates text and code. More pieces of code would then suggest higher quality.⁴ The second measure exploits the fact that authors of the questions can *accept* one of the answers they receive if they consider it satisfactory. This action is not mandatory, so it can reliably indicate whether the questioner was able to solve his problem with the information received. I then measure how much the answerer was incentivised by exploiting the *bounties* website feature. Stack Overflow users can auction reputation points (i.e., virtual rewards) on given questions. In other words, they can reward the author of an acceptable answer. A higher number of points at stake implies more substantial incentives.

The analysis develops two main empirical strategies. The first strategy aims to estimate how lower language barriers affect communication quality, whereas the second strategy aims to test whether new users provide lower-quality content compared with existing users. More precisely, the first empirical analysis studies how the availability of old joiners' native languages impacts their contributions' quality. It relies on a staggered difference-in-difference approach, where a user becomes treated when the platform implements the site in the user's native language. I execute a regression analysis at the answer level with communication quality as the dependent variable, the availability of the user's native language website as a treatment dummy, and time and user fixed effects. I use the estimation technique developed by Borusyak et al. (2024) and compare the results with the more standard two-way fixed effects (TWFE) approach.⁵ I study the complementarities and heterogeneity of this effect by interacting the treatment dummy with different levels of relevant variables. The second empirical strategy aims to compare the answers of new joiners with those of old joiners on the non-English sites. By exploiting the fact that questions can receive multiple answers, this strategy consists of a question fixed effects ordinary least squares (OLS) regression, where answer quality is the dependent variable and different dummy variables identify whether the author is a new joiner or an old joiner. I also include order-of-publication fixed effects and control for the amount of time the user has been registered on the platform to capture the experience of the platform's functioning.

This paper shows that the trade-off is confirmed in the data. The introduction of native language websites increases old joiners' communication quality by 21.4% overall, with quality in native-language answers being 56.5% higher than in pretreatment English answers. The overall effect increases to 118% if users are highly incentivised, and to more than 30% when the question's quality is in the top quartile. Answers are also 17.2% more likely to be *accepted* by the questioner in this case, suggesting that higher quality leads to more effective communication. The estimates suggest substantial heterogeneity:

users who are mainly active in English after treatment have close to a null effect, whereas users who are primarily active in their native language increase answer quality by up to 76.2%. By contrast, the effect is relatively homogeneous across different levels of pretreatment participation and across non-English languages. These results suggest that the platform benefits from allowing communication in multiple languages. Nevertheless, for questions in the native language, answers by new joiners have 0.46 standard deviations lower quality compared with old joiners and 13.8 percentage points lower probability of being accepted. Although other reasons may explain why new joiners do not participate on the English site, the substantial difference in answer quality suggests that the availability of sites in the users' native languages allows for the participation of users with lower expertise.

Overall, knowledge platforms that target quality highly benefit from multiple languages as nonnative English users increase the quality and effectiveness of contributions. Nevertheless, these benefits are reduced when the nonnative English users are very few, have a low cost of using English, or have low expertise. In this case, from an efficiency standpoint, a single language (or a limited number of languages) is preferable.⁶

Although this trade-off is particularly important for knowledge platforms, it is also relevant to other economic and managerial environments. An example is national states (Ginsburgh and Weber 2011) that can impose a single language, like in France (Blanc and Kubo 2024), or maintain language diversity, as in Spain. Firms provide other examples. For instance, many firms need to choose between common and specialised "languages" across divisions (Crémer et al. 2007) and evaluate the trade-offs of multinational teams with language frictions (Chen et al. 2006, Tenzer et al. 2014). Finally, this trade-off is relevant in international trade, where a common language is necessary to find agreements but language costs can prevent efficient interactions (Melitz 2008, Lohmann 2011).

To my knowledge, this is the first paper to empirically quantify the role of language costs in communication quality and its outcomes. Some experimental literature has tested communication games with and without communication frictions (Blume et al. 2001, Lafky and Wilson 2020, respectively). The works of McManus (1985), Tainer (1988), Guillouët et al. (2024), and Battiston et al. (2021) also study exogenous communication costs and their effect on outcomes. As in this paper, McManus (1985), Tainer (1988), and Guillouët et al. (2024) investigate the cost of using English and study its impact on wages and productivity. By contrast, Battiston et al. (2021) focus on the communication frictions arising from being unable to talk face-to-face rather than the language itself. These papers do not observe the actual communication but only communication

outcomes, so they do not quantify the changes in communication quality.

The rest of this paper is organised as follows: Section 2 describes the Stack Overflow platform; Section 3 provides intuitions over the main results with empirical facts; Section 4 presents a theoretical framework and predictions; Section 5 describes empirical proxies; Section 6 derives empirical hypotheses; Section 7 provides the empirical analysis and estimates for the effect on old joiners' answer quality; Section 8 provides the empirical analysis and estimates for the difference in answer quality between old joiners and new joiners; Section 9 discusses managerial implications; and Section 10 presents the conclusion of this study.

2. Communication in Q&A Platforms

Question-and-answer websites are online platforms that allow users to ask questions or answer them. Examples of such websites include Stack Overflow, Yahoo! Answers, and Quora.⁷ These platforms are particularly useful for analysing communication as, compared with offline environments, they provide larger samples, richer data, and less personal interactions, suggesting less confounding factors.

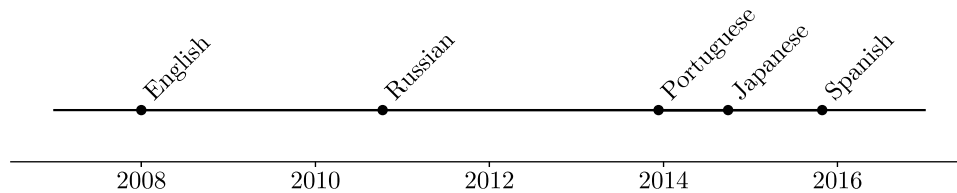
2.1. Stack Overflow

For the empirical investigation, this paper relies on data from Stack Overflow, a question-and-answer website that focuses on topics related to computer programming. Questions may concern, for instance, how to use programming languages for data analysis or software development or how to solve coding bugs. The website has the objective to be the main source of information for all possible problems that programmers may encounter (Spolsky 2008). The key features of the platform are that it is crowd based and free of charge. In other words, any internet user can register on the website for free and ask questions, provide answers, or both. Contributors are not remunerated.⁸

Stack Overflow stands out from other sites because of the size of its welfare impact: many programmers are self-learned, and Stack Overflow provides a large community willing to help. As of June 2021, Stack Overflow receives more than one hundred million monthly visits.⁹ In addition, Stack Overflow offers information seekers content that is easily accessible and searchable via browser search engines. The literature has identified these features as particularly important for productivity gains (Boudreau et al. 2017, Goldfarb and Tucker 2019, Sandvik et al. 2020).

2.2. Languages Used in Stack Overflow

As of June 2021, there are five different Stack Overflow websites, each in a different language, namely, English, Russian, Japanese, Portuguese, and Spanish.¹⁰ Apart

Figure 1. Timeline of the Introduction of the Stack Overflow Websites

from the language, their function is identical. Each website became public at a different time. Stack Overflow was first launched in English in September 2008. The platform was implemented in English as the founders are Americans and the use of English is the norm in the programming community. Nevertheless, they realised that a significant part of the programming community may be unable to access English content. After some discussion, they decided to open Stack Overflow in languages other than English.¹¹

The platform designers chose those four additional languages because large communities of programmers speak them and, at the same time, may not speak English. The introduction of each website followed some beta periods before the rollout of the final version.¹²

Figure 1 shows the timeline of the introduction of the different websites.

2.2.1. The Case of Stack Overflow in Russian. The introduction of Stack Overflow in Russian followed a slightly different process. In 2010, some Russian programmers decided to create a clone of the English Stack Overflow in Russian. They created a website called HashCode, replicating Stack Overflow's features and purpose. Once the company behind Stack Overflow decided to open a version in Russian, it acquired HashCode: on March 31, 2015, all posts from HashCode were imported into the Russian version of Stack Overflow. Formally, the Russian version of Stack Overflow appeared in 2015. The data available include all the HashCode content, implying that the Russian data are available from 2010. Figure 1 reflects this fact and reports the October 10, 2010, as the date for the site launch.

3. Stylised Facts

The introduction of new sites that use different languages mainly affects users who are native speakers of those languages. These users may have or have not contributed in English before their native language became available. I refer to the former as old joiners. In practice, these users participated on the English site before their native language became available and then participated in their native language once available. If the availability of the native language defines a treatment, old joiners are *not-yet-treated* when they can only use English and *treated* afterward. The researcher can

observe their contributions before and after treatment. On the contrary, this is not possible with users who have not contributed to the English site before their native language was introduced. I refer to these users as new joiners on the condition that they had joined the platform before their native language became available.¹³

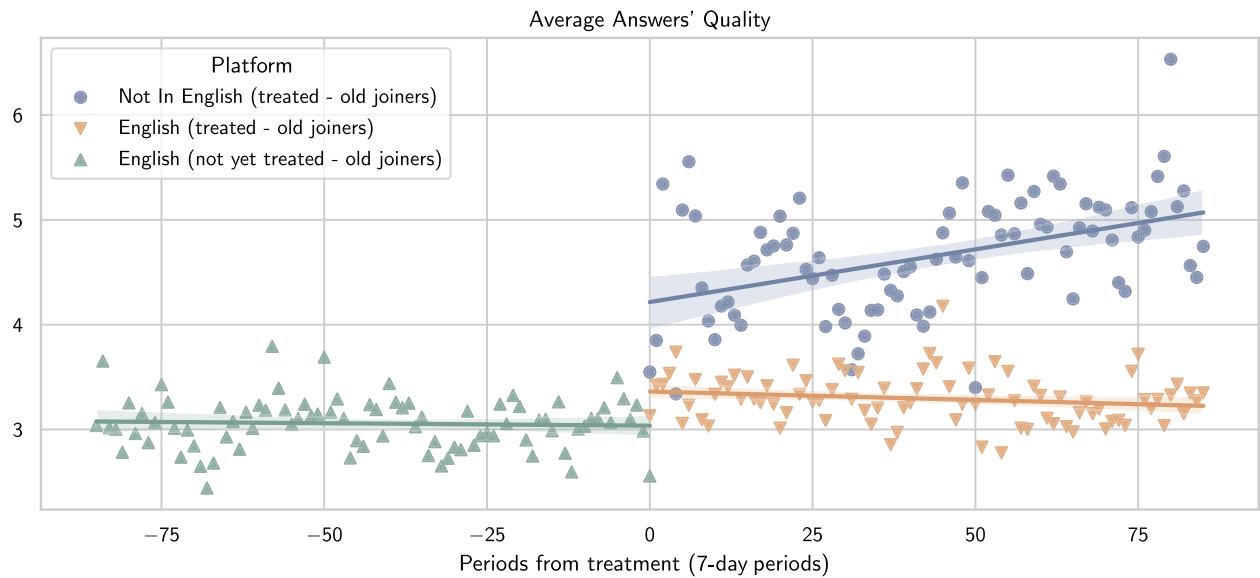
Intuitively, the platform, by allowing users to communicate in their native languages, decreases their cost of participation. On the one hand, this may allow old joiners to increase their contribution quality. On the other hand, it may allow new joiners to start contributing. The latter effect may be positive for the platform as it allows the community size to grow; however, it may have negative implications on content quality if new joiners are poor contributors.

This paper studies these possible effects on quality in detail. The raw data already provide suggestive evidence of the quality trade-off faced by the platform. Figure 2 plots the old joiners' average answer quality based on the language used before and after treatment. Quality is measured as the number of separate snippets of code appearing in the text of the answer. The figure shows that old joiners' answers in the native language have a higher quality compared with those written in English, suggesting that the platform faces an increase in content quality. However, this beneficial effect may be compensated by lower-quality content from new participants. Figure 3 augments Figure 2 by adding the new joiners' average answer quality. It shows that, on average, new joiners contribute lower-quality answers than old joiners. A possible mechanism driving this effect is that language costs act as screening devices for knowledgeable users, as the other users find it too costly to participate. By removing language barriers, the costs are reduced, allowing inexperienced users to contribute low-quality content. This paper provides some theoretical insights to explain these mechanisms, derives testable hypotheses, and investigates them empirically.

4. Theoretical Illustration

In this section, I present a simple framework to provide insights into how the cost of language and other factors affect communication quality between a questioner and an answerer on Stack Overflow. This section aims to guide the empirical analysis and the interpretation of the results in showing how answer quality depends on the question's quality, the incentives between sender and

Figure 2. (Color online) Average Answer Quality Before and After the Introduction of the Native Language Websites

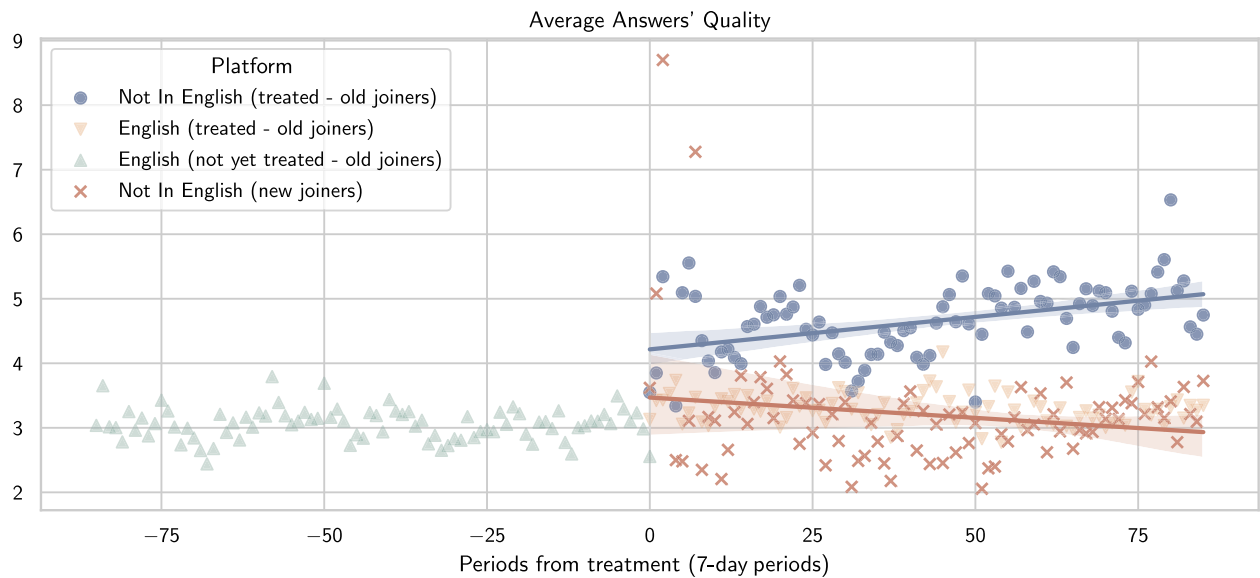


Notes. Each point represents the average number of pieces of code that old joiners included in their answers in a seven-day period. The averages are based on the language used and whether the answer was posted before or after the introduction of the users' native language. Not all old joiners participated in each seven-day period.

receiver, and the answerers' expertise and cost of language, where the latter is the critical variable of interest. In the context of this paper, communication quality is a joint measure of the clarity and informativeness of the information transmitted and directly impacts the degree to which the receiver understands the information. Communication quality depends on both the question's

and answer's quality, which are the outcomes of the transformation of inputs (e.g., effort, time) in the texts of the question and the answer. The framework does not model these processes. Instead, it sets question quality and answer quality as the choice variables, implicitly assuming that the users can manipulate those inputs to achieve a desired level of quality.¹⁴

Figure 3. (Color online) Average Answer Quality of Old Joiners and New Joiners Before and After the Introduction of the Native Language Websites



Notes. Each point represents the average number of pieces of code that old joiners and new joiners included in their answers in a seven-day period. The averages are based on the language used and whether the answer was posted before or after the introduction of the users' native language. Not all old joiners and new joiners participated in each seven-day period.

The framework simplifies and adapts the model proposed by Calvó-Armengol et al. (2015) to pairwise communication and unilateral information transmission.¹⁵ Simplicity comes at the cost of several limitations. In particular, in the model, pairwise communication is independent of other communicating pairs and the answerers' communication choices do not depend on other answerers. In addition, the framework focuses on modelling a single interaction and does not provide insights into how communication costs affect the quantity of answers provided.

These assumptions can be partly justified by the fact that the framework focuses on explaining quality rather than participation intensity and the realisation of quality is conditional on participation. Nevertheless, under these assumptions, the model abstracts from factors that may affect quality choices. These factors include the competition arising from other possible answerers and the potential nonrandom distribution of the questioner's characteristics and the question's topic. Section B.5 in the Online Appendix discusses the issue in more detail, whereas the empirical analyses address these potentially confounding factors.

4.1. Model Setup

Let Bob be a programmer who needs to understand how to implement some features in his software. He decides to ask about his problem to the Stack Overflow community as, otherwise, he cannot proceed with his project. Alice is a community member who sometimes answers questions on the platform. Alice defines her strategy of communication quality before observing the information that Bob needs.¹⁶

Let the information that solves Bob's problem be θ . Both Bob and Alice know its ex ante distribution:

$$\theta \sim \mathcal{N}\left(0, \frac{1}{s}\right).$$

Here, s represents the precision of the prior: if s is high, Bob cannot be too wrong even if he picks a solution at random; if s is low, he may make major mistakes.¹⁷

First, Bob writes the question by choosing its quality Φ_Q . He makes the quality choice by minimising the loss function

$$U_Q = \mathbb{E}[-((a^* - \theta)^2 + C_Q^2 \Phi_Q)],$$

where $a^* \in (-\infty, \infty)$ is the solution that he will pick in the last stage to solve his problem. Thus, the first argument represents his disutility from how different the solution he picks is from the correct solution. The term C_Q is his cost of communication.¹⁸

Second, Alice reads Bob's question: based on her communication strategy, she replies by choosing the answer's quality Φ_A . Alice cares about Bob choosing the right solution to a degree $\gamma \in [0, 1]$. The term γ

determines how much she is inclined to help Bob, which may depend on platform-level incentives (e.g., virtual rewards for providing high-quality answers) or her personal preferences, like empathy and altruism. For what follows, and for simplicity, I will refer to γ as the degree of incentives.¹⁹ She then chooses the answer's quality to minimise the loss function:

$$U_A = \mathbb{E}[-(\gamma(a^* - \theta)^2 + C_A^2 \Phi_A)],$$

where C_A is her cost of communication.

I assume that the cost of communication is a function of the cost of using a given language (λ) and the expertise in the subject (k). Specifically,

$$C_Q = \frac{\lambda_Q}{k_Q} \quad \text{and} \quad C_A = \frac{\lambda_A}{k_A}$$

for Bob and Alice, respectively. This functional form captures the fact that with low proficiency in the language used, users have a high cost of communication. Nevertheless, expertise and familiarity with the topic alleviate that cost.

Once the question and the answer are published, Bob receives the information m , which represents a noisy version of the correct solution

$$m = \theta + \varepsilon + \eta,$$

where the noise terms ε and η are stochastically independent from θ and each other and shrink in the question's and answer's quality. More precisely,

$$\varepsilon \sim \mathcal{N}\left(0, \frac{1}{\Phi_Q}\right); \quad \eta \sim \mathcal{N}\left(0, \frac{1}{\Phi_A}\right).$$

When the quality of the question is higher, the variance of the noise ε is lower and the message is more precise. Similarly, the noise coming from the answer shrinks when the answer's quality is higher.

Finally, Bob uses m to update his prior over the correct solution and picks the optimal solution a^* .

4.2. Optimal Choice of Answer's Quality

For a given question, what is Alice's optimal choice of her answer's quality?²⁰ Proceeding backward, given the received message m , Bob selects the action a^* such that

$$a^* \equiv \arg \max_a \mathbb{E}[-((a - \theta)^2 + C_Q^2 \Phi_Q) | m].$$

By Bayesian updating, the optimal action is then given by

$$a^* = \mathbb{E}[\theta | m] = \beta m \quad \text{with} \quad \beta \equiv \frac{\Phi_Q \Phi_A}{\Phi_Q \Phi_A + \Phi_Q s + \Phi_A s}. \quad (1)$$

In words, Bob weighs the message by its expected informativeness.

Anticipating Bob's action, Alice solves the following:

$$\max_{\Phi_A \geq 0} \mathbb{E}[-(\gamma(a - \theta)^2 + C_A^2 \Phi_A)],$$

and her best response is then given by

$$BR_A(\Phi_Q) = \frac{\Phi_Q(\sqrt{\gamma}k_A - s\lambda_A)}{\lambda_A(\Phi_Q + s)}. \quad (2)$$

4.3. Implications of the Model

This section derives theoretical predictions and testable implications for both old joiners who contributed in English before their native language became available and new joiners who did not. This is achieved by assuming that all answerers participating in the platform follow the same decision process as Alice.

To formally define old joiners and new joiners in the context of the theoretical framework, note that Alice contributes an answer if

$$\sqrt{\gamma}k_A > s\lambda_A. \quad (3)$$

This condition says that Alice contributes an answer if she sufficiently internalises Bob's payoff (i.e., γ is high enough), if Bob is sufficiently confused ex ante about the correct solution (i.e., the precision of the prior, s , is low enough), if her expertise is sufficiently high (i.e., k_A is high enough), and if her cost of using the language is sufficiently low (i.e., λ_A is low enough).

Assuming that Spanish is Alice's native language, it follows that Alice is part of the old joiners if the condition holds for both the English and Spanish sites. Conversely, she is part of the new joiners if the condition holds for the Spanish site but not for the English one.

4.3.1. Implications for the Old Joiners: Comparative Statics on the Cost of Language. Equation (2) shows how answer quality depends on the cost of language. By taking the partial derivative with respect to the level of cost of language λ_A , we obtain

$$\frac{\partial BR_A(\Phi_Q)}{\partial \lambda_A} = -\frac{\Phi_Q \sqrt{\gamma}k_A}{\lambda_A^2(\Phi_Q + s)} < 0, \quad (4)$$

which suggest that higher costs of language induce lower answer quality. The model provides additional predictions. The cross derivatives show that after a drop in the cost of language, the quality increase is larger for higher question quality and higher degree of incentives:

$$\frac{\partial BR_A(\Phi_Q)}{\partial \lambda_A \partial \Phi_Q} = -\frac{\sqrt{\gamma}k_A s}{\lambda_A^2(\Phi_Q + s)^2} < 0, \quad (5)$$

$$\frac{\partial BR_A(\Phi_Q)}{\partial \lambda_A \partial \gamma} = -\frac{\Phi_Q k_A}{2\lambda_A^2(\Phi_Q + s)\sqrt{\gamma}} < 0. \quad (6)$$

These comparative statics lead to the following proposition.

Proposition 1. *When users can answer in their native language,*

1. *their answer quality increases,*
2. *the effect increases when the quality of the question is higher, and*
3. *the effect increases when the answerer is more incentivised.*

Proposition 1 relies on the assumption that users active in non-English languages are native speakers of those languages and more proficient in those languages than in English.²¹ Point 1 directly follows from Equation (4), which establishes a negative relationship between the answer's quality and the author's cost of using the language. The use of the native language rather than English induces a decrease in the cost of language and allow users to provide higher quality answers.²² Point 2 follows from Equations (5), which show that the effect of the cost of language on the answer's quality increases with the question's quality. This effect is due to the fact that the model features complementarities between the questioner's and the answerer's quality choices. This is not imposed as an assumption but emerges endogenously. Indeed, question quality and answer quality are complements in determining the informativeness of the communication, as shown in Equation (1). Such complementarity is then anticipated and internalised by the answerer's decision. Finally, point 3 follows from Equation (6), which shows that the degree to which the answerer cares about the questioner's utility augments the effect of a change in the cost of language. This effect emerges due to complementarity in the answerer's quality choice between the incentives and the cost of language.

4.3.2. Implications for the New Joiners. Condition (3) states that, with the incentives (γ) and the prior's precision (s) fixed, Alice's participation is conditional to a level of expertise high enough relative to the cost of language. If she does not have a sufficiently high level of expertise, she does not participate in English but may be able to participate in her native language because lower language costs loosen the condition to participate.

Formally, consider a question on the topic ω and let j index possible answerers for that question. Assume that the answerers differ only on their expertise on ω , k_{A_j} . Define λ_A as their cost of using English and λ'_A as their cost of using the native language, with $\lambda'_A < \lambda_A$. Let $f(k_A)$ be the probability distribution of expertise on ω across answerers and assume it is the same on both the English and native language sites. We can then say that a user is part of the old joiners (*OJ*) or the new

joiners (NJ) based on the level of expertise:

$$\begin{aligned} j \in OJ & \text{ if } k_{A_j} > \frac{s\lambda_A}{\sqrt{\gamma}}, \\ j \in NJ & \text{ if } \frac{s\lambda'_A}{\sqrt{\gamma}} < k_{A_j} < \frac{s\lambda_A}{\sqrt{\gamma}}. \end{aligned}$$

It then follows that, for any j, j' if $j \in OJ$ and $j' \in NJ$,

$$\mathbb{E}[k_{A_j} | \omega] \geq \mathbb{E}[k_{A_{j'}} | \omega]. \quad (7)$$

In other words, for a given question's topic, answerers who are old joiners have higher expertise than answerers who are new joiners. Note that this result generalises to the case in which answerers are heterogeneous on their cost of using English.²³

Note that from the best response function reported in Equation (2), the choice of the answer's quality is linear on the answerer's expertise. From Equation (7), it follows that, for any j, j' if $j \in OJ$ and $j' \in NJ$,

$$\frac{\Phi_Q(\sqrt{\gamma}\mathbb{E}[k_{A_j} | \omega] - s\lambda'_A)}{\lambda'_A(\Phi_Q + s)} \geq \frac{\Phi_Q(\sqrt{\gamma}\mathbb{E}[k_{A_{j'}} | \omega] - s\lambda'_A)}{\lambda'_A(\Phi_Q + s)}.$$

This result implies the following proposition.

Proposition 2. *On average, for a given topic, answers from old joiners have higher quality than answers from new joiners.*

5. Empirical Proxies

To test empirically the theoretical predictions, it is necessary to identify observable proxies for the model's variables.

5.1. Measuring the Answers' Quality

To proxy for the answers' quality, this paper uses the number of separate snippets of code in the text of the answer. I refer to this variable as *numCodes*. More precisely, each answer is an HTML script. To include code snippets in their answers, users add *code* sections (i.e., `<code> ... </code>`) to make the code appear in a separate box with a different colour background. The box mimics a programming/statistical software console and makes the code more readable. The proxy of quality is then defined as the number of code sections in the answer. The intuition behind this measure is that a typical answer about programming includes some textual explanations and some code snippets to illustrate the solution. The presence of multiple snippets may indicate that the author is either providing several pieces of information (higher informativeness) or explaining one piece of information more clearly with a step-by-step procedure (higher clarity). In both cases, more snippets suggest higher answer quality.²⁴

Compared with alternative proxies, this measure has two main advantages. First, it is not directly based on the language itself, which would create a systematic

difference across sites by construction. This issue would affect measures based on text or other text characteristics. Second, the number of snippets in an answer is a direct choice of the answerer, as the model assumes it to be.

An alternative way to measure quality relies on community feedback (e.g., the number of *likes* in the question). Although measures of this type capture communication quality as a whole rather than the answers' quality, they provide information on the value of the answer to the community. I replicate the analysis with a measure of quality based on whether the questioner has accepted the answer. Indeed, questioners can accept the answer that solved their problem, if any. I refer to this dummy variable as *Is Best Answer*.²⁵

5.2. Measuring the Questions' Quality

To measure the questions' quality, I adopt the same strategy as for the answers and use the number of separate snippets of code appearing in the text of the question as a proxy. The intuition is that a question with code is clearer as it provides context for the problem and a direct way for the answerer to reproduce the error. More pieces of code provide more examples, different attempts, and so on, all of which can help the answerer clearly understand the question.²⁶

5.3. Measuring Incentives

To capture the degree of incentives, I exploit the so-called *bounty* system implemented by Stack Overflow. The platform allows the questioner to auction a certain amount of his reputation points (i.e., virtual points that the questioner has accumulated through on-site activity) on a question. The questioner commits to allocating these points to the user who provides a satisfactory answer. Once the questioner auctions the points, these are removed from his account, although the questioner is not obliged to designate a recipient. The points may remain unallocated if the questioner considers all received answers unworthy.²⁷ This feature of Stack Overflow represents a form of virtual payment with which the questioner can provide extra incentives. A larger bounty amount implies a more significant stake in the question and a stronger incentive for the answerer.

More precisely, consider an answer i to a question $q(i)$. The proxy for incentives is then the bounty amount auctioned on question $q(i)$ and not yet assigned when answer i is published.

6. Empirical Predictions

This section presents the empirical counterpart of the theoretical predictions, representing the hypotheses that the empirical analysis addresses. The two sets of

empirical predictions correspond to Propositions 1 and 2, respectively.

6.1. Treatment Effect on the Quality of Old Joiners' Answers

Proposition 1 states that, everything else equal, the answer quality of old joiners increases once they are able to use their native language rather than English only, and that the effect is higher with higher question quality and more incentives for the answerer.

Given the main empirical measures described in Section 5, the derived empirical hypotheses are as follows.

Hypothesis 1. *Old joiners include more snippets of code in their answers when they use their native language compared with when they use English.*

Hypothesis 2. *The increase in answer quality due to the use of the native language instead of English is larger when the question's quality is higher.*

Hypothesis 3. *The increase in answer quality due to the use of the native language instead of English is larger when the active bounties auctioned on the question have a higher value.*

6.2. Language Costs as Screening Devices

Proposition 2 shows that, everything else equal, new joiners are on average less expert than old joiners and consequently write lower-quality answers.

Because the researcher does not observe the answerers' expertise, it is not possible to identify new joiners as those who did not participate on the English site strictly because of insufficient expertise. Empirically, I define new joiners as users who (1) registered on the English site before their native language became available, (2) did not contribute in English, and (3) contributed in their native language once it became available. This empirical definition rules out the possibility that these users did not contribute to the English site because they were unaware of the platform, although other explanations

remain possible. It follows that the empirical analysis studies the differences in answer quality that are at least partially attributable to differences in expertise.²⁸

The testable hypothesis is stated as follows.

Hypothesis 4. *On average, for a question written in their native language, users already active in English before treatment (old joiners) write answers with more pieces of code compared with users who registered before treatment but are not active in English (new joiners).*

7. The Effect on the Quality of Old Joiners' Answers

This section empirically tests Hypotheses 1, 2, and 3. It studies whether the introduction of additional languages on the platform has affected the quality of old joiners' answers.

7.1. Data

The data include all answers of users participating on both the English and a non-English site (the old joiners, or *treatment* group) and all the answers of a random sample of users who contribute only to the English site (*never-treated* group).²⁹

The treatment occurs if the user has the possibility to write in the native language. If a user is part of the old joiners, she is *treated* if her native language is available (i.e., if the site using her native language has already launched) and *not-yet-treated* if her native language is not yet available. For the sake of simplicity, I say that an answer is *treated* if its author is treated, *not-yet-treated* if its author is not yet treated, and *never treated* otherwise. Note that all answers written after the launch of the native language site are treated, regardless of whether they are in the native language or English.³⁰

Table 1 reports the number of observations by the language and treatment status of the author. The data are right-censored at the end of August 2017, as those were the latest available data at the time of the data retrieval.³¹

Table 1. Number of Observations in the Baseline Sample and Subsamples Based on the Language and Treatment Status of the Author

Sample	Number of answers	Number of authors	Number of answers per author				
			Mean	Std. dev.	Min	Median	Max
English (never treated)	37,049	2,661	13.92	63.13	1.00	2.00	2,175.00
English (not yet treated)	128,982	2,678	48.16	184.82	1.00	5.00	2,848.00
English (treated)	100,537	2,087	48.17	224.25	1.00	7.00	4,894.00
Not in English (treated)	57,282	2,678	21.39	134.73	1.00	2.00	3,759.00
Full sample	323,850	5,339	60.66	266.49	1.00	7.00	6,596.00

Notes. The definitions of the subsamples of answers are as follows: "English (not yet treated)" is the set of answers that users in the treatment group write before treatment; "English (treated)" is the set of answers that users in the treatment group write after treatment in English; "Not in English (treated)" is the set of answers that users in the treatment group write after treatment not in English; and "English (never treated)" is the set of answers written by users in the never-treated group. By construction, the same users compose the English (not yet treated) and Not in English (treated) subsamples. Because some users stop participating in English after treatment, the set of users in the English (treated) group is smaller but still contained in the treatment group sample. Std. dev., standard deviation.

The observed variables are at the answer level and represent either answer characteristics (quality, number of likes, whether the answer has been accepted) or characteristics of the question that the answer is answering. The latter characteristics include quality, the bounty amount, the number of answers and views received, and the characteristics of the questioner when available, including whether the questioner shares the language with the answerer and has uploaded a profile picture. Table 2 reports summary statistics of the numeric variables by treatment status of the answer’s author and the language in which the answer is written.

7.2. Empirical Strategy

The empirical strategy exploits the sequential introduction of websites in Russian, Portuguese, Spanish, and Japanese in addition to the English site. The ability to track users’ contributions across sites allows for measuring changes in quality choices for each user before and after their native language site became available, and comparing them with the never-treated users. In this setting, I can identify the effect of the availability of the native language on answers’ quality (the treatment effect on the treated) using a difference-in-difference design with staggered treatment. Section C.1 in the Online Appendix provides details on the identifying

Table 2. Summary Statistics at the Answer Level for the Baseline Sample by the Site’s Language and the Language and Treatment Status of the Author

Sample	Variable	Mean	Std. dev.	Min	Median	Max
English (never treated)	<i>Answer’s numCodes</i>	2.01	2.69	0.0	1.0	67.0
	<i>Answer’s Score</i>	3.62	20.55	−13.0	1.0	1,264.0
	<i>Is Best Answer</i>	0.31	0.46	0.0	0.0	1.0
	<i>Question’s numCodes</i>	2.13	2.70	0.0	1.0	41.0
	<i>Bounty Amount</i>	1.88	21.63	0.0	0.0	1,200.0
	<i>Question’s Num. Answers</i>	3.30	4.64	1.0	2.0	407.0
	<i>Question’s Num. Views</i>	15,062.10	93,963.64	9.0	1,090.0	4,642,034.0
	<i>Same Native Language (SNL)</i>	—	—	—	—	—
	<i>SNL and Manual Profile Picture</i>	—	—	—	—	—
English (not yet treated)	<i>Answer’s numCodes</i>	2.74	4.02	0.0	2.0	284.0
	<i>Answer’s Score</i>	3.45	18.60	−16.0	1.0	1,902.0
	<i>Is Best Answer</i>	0.36	0.48	0.0	0.0	1.0
	<i>Question’s numCodes</i>	2.23	2.77	0.0	1.0	111.0
	<i>Bounty Amount</i>	1.16	14.70	0.0	0.0	700.0
	<i>Question’s Num. Answers</i>	3.28	5.95	0.0	2.0	518.0
	<i>Question’s Num. Views</i>	10,618.68	72,795.79	7.0	1,003.0	7,465,142.0
	<i>Same Native Language (SNL)</i>	0.02	0.12	0.0	0.0	1.0
	<i>SNL and Manual Profile Picture</i>	0.01	0.07	0.0	0.0	1.0
English (treated)	<i>Answer’s numCodes</i>	3.55	4.65	0.0	2.0	153.0
	<i>Answer’s Score</i>	2.33	11.76	−16.0	1.0	1,068.0
	<i>Is Best Answer</i>	0.41	0.49	0.0	0.0	1.0
	<i>Question’s numCodes</i>	2.69	3.07	0.0	2.0	71.0
	<i>Bounty Amount</i>	1.14	14.33	0.0	0.0	800.0
	<i>Question’s Num. Answers</i>	2.52	2.86	0.0	2.0	116.0
	<i>Question’s Num. Views</i>	9,058.16	87,796.96	8.0	603.0	8,671,208.0
	<i>Same Native Language (SNL)</i>	0.02	0.13	0.0	0.0	1.0
	<i>SNL and Manual Profile Picture</i>	0.01	0.09	0.0	0.0	1.0
Not in English (treated)	<i>Answer’s numCodes</i>	4.64	6.15	0.0	3.0	186.0
	<i>Answer’s Score</i>	2.91	4.57	−5.0	2.0	256.0
	<i>Is Best Answer</i>	0.47	0.50	0.0	0.0	1.0
	<i>Question’s numCodes</i>	2.56	3.05	0.0	2.0	87.0
	<i>Bounty Amount</i>	0.95	12.64	0.0	0.0	800.0
	<i>Question’s Num. Answers</i>	2.01	1.30	1.0	2.0	27.0
	<i>Question’s Num. Views</i>	953.91	3,382.53	5.0	177.0	114,994.0
	<i>Same Native Language (SNL)</i>	1.00	0.00	1.0	1.0	1.0
	<i>SNL and Manual Profile Picture</i>	0.52	0.50	0.0	1.0	1.0

Notes. *Answer’s numCodes* is the number of pieces of code appearing in the answer; *Answer’s Score* is the number of likes net of dislikes that the answer received; *Is Best Answer* is a dummy equal to one if the questioner accepted the answer as the one solving the problem; *Question’s numCodes* is the number of pieces of code appearing in the question; *Bounty Amount* is the number of points auctioned on the question; *Question’s Num. Answers* is the number of answers that the question has received, including the one of the observation; *Question’s Num. Views* is the number of users that have viewed the question; *Same Native Language* is a dummy equal to one if the author of the answer and the questioner speak the same language; and *Manual Profile Picture* is a dummy equal to one if the questioner both has a personalised profile picture and speaks the same language as the answerer. The *Same Native Language* and *Manual Profile Picture* variables are not observed for the *never treated* group. Std. dev., standard deviation.

assumptions, describes possible violations, and discusses how I address them, including adding control variables in the regressions and through robustness checks.

7.2.1. Estimation Method. The literature has proposed several methods to estimate the average treatment effect on the treated (ATT) in staggered difference-in-difference designs. In this paper, I provide estimates under two approaches. The first is the so-called two-way fixed effects method, which I present as a benchmark because it is the most commonly used method, although I claim it provides biased estimates in this environment. The second is the method proposed by Borusyak et al. (2024).³²

The two-way fixed effect method is an OLS model with individual and time fixed effects. More precisely, the estimating equation is

$$\text{numCodes}_i = \alpha_{j(i)} + \delta_{t(i)} + \beta D_{L(j(i), t(i))} + \mathbf{W}'_i \boldsymbol{\rho} + \varepsilon_i, \quad (8)$$

where i indexes an answer written by the user $j(i)$ at the time $t(i)$; D is a dummy variable taking a value equal to one if the answer i was written when the site in the language L , j 's native language, was available; $\mathbf{W}$ is a vector of control variables; $\alpha_{j(i)}$ is the author fixed effect, and $\delta_{t(i)}$ is the time fixed effect. The coefficient of interest β then captures the ATT. However, the identification relies on the assumption that all treatment effects for all users and periods are the same.³³ This assumption does not hold as the treatment effect should depend on the size of the change in the cost of language, which is different across users if the users are heterogeneous in the cost of using English.

To allow for less restrictive assumptions, the preferred estimation strategy is the method proposed by Borusyak et al. (2024) (BJS hereafter).³⁴ The method builds on the intuition that, under the parallel trends assumption, a model trained with the *not-treated* data (i.e., never-treated and not-yet-treated answers) can predict the potential outcome, that is, the counterfactual choice of treated users if they had not been treated. More precisely, the BJS method comprises three steps:

Step 1. Estimate with OLS a linear model on the never-treated and not-yet-treated answers only:

$$\text{numCodes}_i = \alpha_{j(i)} + \delta_{t(i)} + \mathbf{W}'_i \boldsymbol{\rho} + \varepsilon_i$$

if $j(i)$ not treated at time $t(i)$,

with $\alpha_{j(i)}$ and $\delta_{t(i)}$ being the individual and time fixed effects and $\mathbf{W}_i$ being additional control variables. This step estimates parameters for individual and time fixed effects for the whole set of individuals and time periods.³⁵

Step 2. With the model estimated in Step 1, predict the counterfactual quality choice of treated users had they not been treated. Then, compute the treatment

effect for each treated user's answer as the difference between the observed quality and the predicted counterfactual quality:

$$\begin{aligned} \widehat{\text{numCodes}}_i &= \hat{\alpha}_{j(i)} + \hat{\delta}_{t(i)} + \mathbf{W}'_i \hat{\boldsymbol{\rho}} \quad \text{if } j(i) \text{ treated at time } t(i), \\ \hat{\tau}_i &= \text{numCodes}_i - \widehat{\text{numCodes}}_i \quad \text{if } j(i) \text{ treated at time } t(i). \end{aligned}$$

Step 3. Compute the ATT as the average of all answers' treatment effects. This step can take different forms as the researcher has the flexibility to select the best way to average across treatment effects. The baseline estimates use a simple average with homogeneous weights:

$$\hat{\tau} = \frac{1}{N_{\text{post}} \sum_{i|j(i) \text{ treated at time } t(i)}} \hat{\tau}_i, \quad (9)$$

where N_{post} is the number of answers published by the treated users.³⁶

7.3. The Effect of the Cost of Language on the Answers' Quality

Table 3 reports the estimated treatment effect corresponding to $\hat{\beta}$ in the TWFE specification (Equation (8)) and to $\hat{\tau}$ in the BJS specification (Equation (9)). Different columns include or exclude certain control variables. *QQuality* identifies the proxy for question quality as described in Section 5.2; *Competition* identifies the variables capturing the number of answers in the question thread and the number of views of the question; and *Empathy* identifies the variables capturing whether the questioner is a native speaker of the same language as the answerer and whether the questioner both displays a personalised profile picture and shares the same native language.³⁷

The estimates show that when users can write answers in their native language, their answers have significantly higher quality on average, and questioners are more likely to solve their problems. The first set of estimates refers to specifications with *numCodes* as the dependent variable and quantifies the effect shown in Figure 2. The preferred specification shows that, on average, answers have 0.586 additional pieces of code. Because the pretreatment average for users in the treatment group is 2.74, as described in Table 2 for the sample "English (not yet treated)," the preferred specification in column (8) of Table 3 reports a 21.4% increase in quality. In other words, answers written posttreatment, in either English or users' native languages, have 21.4% higher quality than English pretreatment answers. Section E.1 in the Online Appendix shows that the effect increases to 56.5% when focusing on non-English answers only, whereas Section G.2 in the Online Appendix shows that, in the preferred specification, the effect is not significant when including only English answers. These results suggest that non-English answers drive the overall effect,

Table 3. Effect of the Cost of Language on the Answers' Quality

DV	(1) TWFE	(2) TWFE 1	(3) TWFE 2	(4) TWFE 3	(5) BJS	(6) BJS 1	(7) BJS 2	(8) BJS 3
<i>numCodes</i>	0.378* (0.0929)	0.375* (0.0968)	0.376* (0.0970)	0.217* (0.0601)	0.574*** (0.0445)	0.591*** (0.0426)	0.596*** (0.0400)	0.586*** (0.0629)
<i>Is Best Answer</i>	0.0225** (0.00365)	0.0223** (0.00371)	0.0213** (0.00349)	0.00894 (0.00391)	0.105*** (0.00594)	0.105*** (0.00593)	0.0915*** (0.00512)	0.0624*** (0.00760)
Observations	323,850	322,992	322,992	285,943	323,850	322,919	322,919	204,541
CSE	Nat-lang	Nat-lang	Nat-lang	Nat-lang	Nat-lang	Nat-lang	Nat-lang	Nat-lang
Controls								
<i>QQuality</i>	No	Yes	Yes	Yes	No	Yes	Yes	Yes
<i>Competition</i>	No	No	Yes	Yes	No	No	Yes	Yes
<i>Empathy</i>	No	No	No	Yes	No	No	No	Yes

Notes. Standard errors are in parentheses. The table shows treatment effect estimates where the dependent variable (DV) is the number of pieces of code (*numCodes*) or a dummy equal to one if the answer is accepted (*Is Best Answer*). The estimates correspond to the average treatment effect on the treated and correspond to the parameter $\hat{\beta}$ or $\hat{\tau}$ when the specification adopted is the TWFE or the BJS, respectively. The standard errors are clustered (CSE) at the native language (Nat-lang) level, that is, at the treatment level.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

confirming the theory and Hypothesis 1. The second set of estimates refers to specifications where the dependent variable is a dummy equal to one if the answer has been accepted and zero otherwise (*Is Best Answer*). The preferred specification in column (8) shows that the answers' probability of being accepted increases by 6.2 percentage points. This effect corresponds to a 17.2% increase from the pretreatment average (36%) and suggests that a 21% increase in the answers' informativeness leads to a 17% increase in their effectiveness.

7.4. Complementarities of the Effect with the Questioners' Choices and Incentives

The theoretical predictions (5) and (6) suggest that the positive impact of a decrease in the cost of language is complementary to the questions' quality and the degree of incentives. To test these predictions and the corresponding Hypotheses 2 and 3, I run the same estimation strategy used for the baseline estimate but allow the coefficient of interest to vary across the levels of those variables.

Let c index the categories for each of the levels. The specification for the TWFE estimation method is the following:

$$numCodes_i = \alpha_{j(i)} + \alpha_{t(i)} + \sum_c \beta_c D_{L(j(i), t(i))} \mathbf{1}_{c(i)} + \mathbf{W}'_i \boldsymbol{\gamma} + \varepsilon_i, \quad (10)$$

where $\mathbf{1}_{c(i)}$ is an indicator function taking a value equal to one if the answer i belongs to the level category c . Regarding the BJS estimation method, the only difference is in the last step. After estimating potential outcomes and treatment effects for each treated observation, the ATT results from averaging the treatment effects within each category level rather than across the all treated sample. Formally, the estimated treatment

effect for category c is

$$\hat{\tau}_c = \frac{1}{N_c} \sum_{i|j(i) \text{ treated at time } t(i)} \hat{\tau}_i \mathbf{1}_{c(i)}, \quad (11)$$

where N_c is the number of treated answers that belong to category c .

7.4.1. The Treatment Effect Increases in the Quality of the Question. To test Hypothesis 2, I separately estimate the treatment effect for different levels of question quality. As a proxy for the questions' quality, I use the number of separated snippets of code that the questioner included in the question (the variable *QQuality*). I bin this variable into four levels based on the quartiles of its distribution. This leads the *Low* category to include questions with either zero or one piece of code, the *MediumLow* category questions with two pieces of code, the *MediumHigh* category questions with three pieces of code, and the *High* category questions with more than three and up to 111 pieces of code.

Table 4 reports the estimates. Columns (1)–(3) contain the $\{\hat{\beta}_c\}_{\forall c}$, whereas columns (4)–(6) contain the $\{\hat{\tau}_c\}_{\forall c}$, where c indexes the categories for *QQuality*. Columns (2) and (4) report the results for the specification that includes all the controls. These columns suggest that low-quality questions receive significantly lower-quality answers, whereas the answers' quality is quite homogeneous across other levels of question quality. Nevertheless, removing the question's quality as a control (columns (4) and (6)) provides substantially different estimates and shows that the treatment effect increases with question quality. An intrinsic difference between the questions with zero snippets of code and those with a positive number of pieces of code may drive this change in estimates, as shown in the Online

Table 4. Treatment Effect Estimates by Level of Question Quality

	(1) TWFE	(2) TWFE 2	(3) TWFE 3	(4) BJS	(5) BJS 2	(6) BJS 3
<i>Low</i>	0.121 (0.118)	−0.0394 (0.0749)	−0.307* (0.0705)	0.288*** (0.0626)	0.317*** (0.0781)	0.0697 (0.153)
<i>MediumLow</i>	0.565** (0.0851)	0.418** (0.0562)	0.343** (0.0408)	0.775*** (0.0866)	0.809*** (0.101)	0.736*** (0.0841)
<i>MediumHigh</i>	0.566** (0.0882)	0.415** (0.0485)	0.478*** (0.0291)	0.792*** (0.0692)	0.827*** (0.0910)	0.881*** (0.124)
<i>High</i>	0.596*** (0.0546)	0.423*** (0.0301)	0.984*** (0.0330)	0.896*** (0.0411)	0.824*** (0.0498)	1.328*** (0.242)
Observations	322,992	285,943	285,943	322,919	204,541	204,545
CSE	Nat-lang	Nat-lang	Nat-lang	Nat-lang	Nat-lang	Nat-lang
Controls						
<i>QQuality</i>	Yes	Yes	No	Yes	Yes	No
<i>Competition</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Empathy</i>	No	Yes	Yes	No	Yes	Yes

Notes. Standard errors are in parentheses. The *Low* category includes questions with either zero or one piece of code, the *MediumLow* category questions with two pieces of code, the *MediumHigh* category questions with three pieces of code, and the *High* category questions with more than three and up to 111 pieces of code. The standard errors are clustered (CSE) at the native language (Nat-lang) level, that is, at the treatment level.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Appendix (Section E.5).³⁸ As a consequence, column (6) of Table 4 reports the preferred specification and shows that the treatment effect increases with question quality.

7.4.2. The Treatment Effect Increases in the Amount of Incentives. To test Hypothesis 3, I separately estimate the treatment effect for different levels of incentives. To measure the degree of incentives between the two parties, I use the values of the bounties at stake on the question that the answer addresses, as described in Section 5.3. I discretise the bounty amount into four categories. The *Low* category is just composed of the zero amount, whereas the other three categories are based on the 33rd and 66th quantiles of the distribution of the positive amounts. In practice, the *MediumLow* category includes questions with 50 points at stake, the *MediumHigh* questions with 100 points at stake, and the *High* category questions with 150 to 1,200 points at stake.

The results are reported in Table 5 and show that, on average, the treatment effect is higher when authors are more incentivised, as suggested by the theoretical framework. The estimates show that, in the absence of incentives via bounties, the answers’ quality increases by about 21%. Note that bounties are not very frequent, which explains why this effect is substantially comparable to the baseline effect reported in Section 7.3. The treatment effect increases to 37% when questions have 50 virtual points at stake, to 86% with 100 points at stake, and to 118% with 150 or more points at stake.

8. Language Costs as Screening Devices

This section aims to test the empirical Hypothesis 4, which states that new joiners contribute lower-quality answers compared with old joiners.

8.1. Data

The data include all answers that satisfy two conditions: (1) they are published in non-English languages, and (2) they answer questions that received at least two answers at the time of the data retrieval. This leads to a final sample of 212,439 answers.³⁹

A key categorical variable splits the sample based on the answerers’ participation on the English site. Whereas some users never registered on the English site, others registered before their native language became available, and others registered after it became available. Moreover, the users who registered may or may not have contributed in English. Table 6 summarises these possible combinations and reports the sample size for each category. The main groups of interest are those identifying new joiners and old joiners. The former are answerers who registered on the English site before their native language became available but did not contribute any answer in English before or after the native language site became available. The latter are answerers who registered and contributed answers on the English site before their native language became available.

8.2. Empirical Strategy

By exploiting the fact that each question can receive multiple answers, the empirical approach compares the answers’ quality between old joiners and new

Table 5. Treatment Effect Estimates by Level of Incentive

	(1) TWFE	(2) TWFE 2	(3) BJS	(4) BJS 2
<i>Low</i>	0.360* (0.0960)	0.202* (0.0586)	0.578*** (0.0400)	0.574*** (0.0631)
<i>MediumLow</i>	1.220* (0.273)	1.043* (0.255)	1.511*** (0.198)	1.007*** (0.191)
<i>MediumHigh</i>	2.298* (0.802)	2.133 (0.857)	2.728*** (0.515)	2.366*** (0.436)
<i>High</i>	3.034*** (0.266)	2.876** (0.308)	3.431*** (0.463)	3.234*** (0.394)
Observations	322,992	285,943	322,919	204,541
CSE	Nat-lang	Nat-lang	Nat-lang	Nat-lang
Controls				
<i>QQuality</i>	Yes	Yes	Yes	Yes
<i>Competition</i>	Yes	Yes	Yes	Yes
<i>Empathy</i>	No	Yes	No	Yes

Notes. Standard errors are in parentheses. The *Low* category includes questions with zero points at stake, the *MediumLow* questions with 50 points at stake, the *MediumHigh* questions with 100 points at stake, and the *High* category questions with 150 to 1,200 points at stake. The standard errors are clustered (CSE) at the native language (Nat-lang) level, that is, at the treatment level.
* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

joiners within each question’s thread. The empirical model corresponds to an OLS regression with the user-group (old joiners versus new joiners) fixed effects, question fixed effects, order-of-publication fixed effects, and a control variable for seniority on the platform. This design allows us to make the quality comparison conditioning on a given topic and the characteristics of the questioner. In addition, the order-of-publication fixed effects control for the fact that the answers arriving second or later in the thread may systematically have less content as they just complement the information already available in the thread. Finally, the inclusion of the amount of time that the answerer has been registered on the platform as a control variable allows us to account for different degrees of experience. Indeed, if new joiners generally registered later, this may induce lower-quality answers as they are less experienced with the platform.

Note that because the analysis uses data from only the non-English sites, the concerns about systematic differences between the English and non-English sites

described in Section B.5 in the Online Appendix are not applicable.

8.2.1. Estimation Method. The unit of analysis is an answer i . Let $j(i)$ denote the author of the answer, $q(i)$ the question it is answering, and $r(i)$ the order by date of publication such that, for instance, $r(i) = 1$ if i is the first answer published for the question $q(i)$. Let G be the set of user types based on whether they registered and eventually participated on the English site. Finally, let $t_{j(i),i}$ be the number of days since the first registration of user j on the Stack Overflow platform until the publication of the answer i . This variable aims to capture the users’ experience and knowledge of the platform’s functioning. The empirical specification is the following:

$$numCodes_i = \alpha_{q(i)} + \delta_{r(i)} + \sum_{g \in G} \beta_g D_{g(j(i))} + \zeta t_{j(i),i} + \varepsilon_i,$$

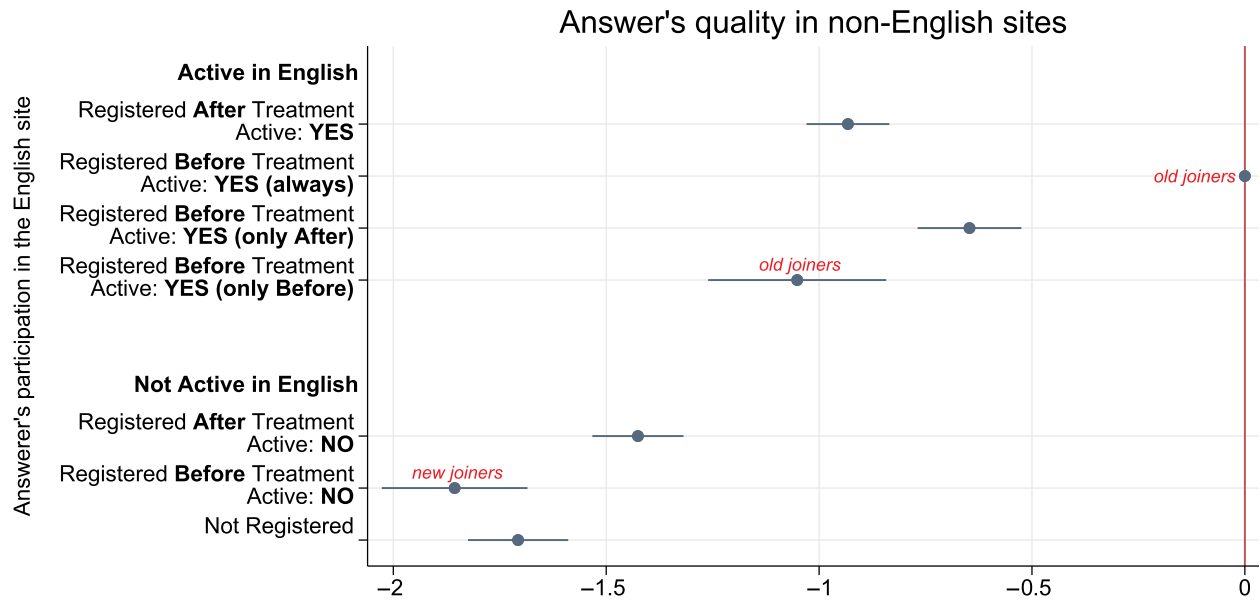
where α and δ are the question and order-of-publication fixed effects, whereas the β s are fixed effect estimates for each group of users.

Table 6. Number of Answers in the Sample by Author Group

Registered on the English site	Active on the English site			Not active
	Only BEFORE	Only AFTER	Both BEFORE and AFTER	
BEFORE	2,589 ^a	9,517	32,860 ^a	4,373 ^b
AFTER		78,471		50,046
Not registered				34,583
Total			212,439	

Notes. The groups are based on participation on the English site. BEFORE and AFTER refer to the treatment date.
^aAnswers of authors considered old joiners are marked with the symbol.
^bAnswers of authors considered new joiners are marked with the symbol.

Figure 4. (Color online) Difference in Answer Quality Between New Joiners and Old Joiners



Notes. Estimates for user group fixed effects on answer quality. The dependent variable is the answers' quality and is proxied by the number of pieces of code included in the answer. The baseline group is users who contributed in English both before and after treatment.

8.3. Results

Figure 4 reports the estimates of $\{\beta_g\}_{g \in G}$ and the corresponding confidence intervals. It shows that, for a given question, new joiners produce significantly lower-quality answers compared with old joiners, as the theory suggests.⁴⁰ This result highlights the potential role of language costs as barriers to participation for low-expertise users.

9. Managerial Implications

Similar to other knowledge platforms, Stack Overflow needs to decide whether to allow contributions in multiple languages. Several dimensions of reasoning can affect the decision, including ethics, inclusivity, stakeholder preferences, and efficiency. Given the contribution of this paper, this discussion focuses on the latter, particularly on the trade-offs concerning the quality of contributions.⁴¹

By introducing multiple languages, the platform alleviates communication costs for users native to those languages. First, it allows users who were already contributing to increase their answers' quality. Second, it allows some of the users who were not participating to start contributing.

The first effect is substantial. On average, when they can use their native language in addition to English, users include 0.586 additional pieces of code, corresponding to a 21.4% increase from the pretreatment average. This implies a substantial increase in information clarity and the amount of information provided. The increase in quality is most likely beneficial to the platform and may also increase consumer satisfaction,

as questioners are more likely to find solutions to their problems. The probability that an answer is accepted increases by 6.2 percentage points when the answerers are able to use their native language. This means that answers are 17.2% more likely to solve the questioners' problems, suggesting that the increase in quality leads to more effective and valuable communication.

However, the second effect may offset the overall benefits of the platform. The participants who start contributing because of the availability of their native language tend to provide lower-quality answers. Their answers have between 0.94 and 1.96 fewer pieces of code than the answers of users who are active in English. These values correspond to a lower quality of 0.22 to 0.46 standard deviations, respectively. In addition, the probability that the questioners accept the answer as a solution to their question is 7.3 to 13.8 percentage points lower. As suggested in this paper, a possible mechanism is that language barriers prevent the participation of low-expertise users. By reducing the cost of communication, the platform loses a screening device and may experience an inflow of lower-quality content.

These results suggest that the platform, when deciding whether to implement additional languages, should take into consideration the sizes of the affected communities. In addition, the results underline how the effective implementation of new languages should be careful about the design of incentives for both questioners and answerers. Indeed, the increase in answer quality is nearly twice as large when the question's quality is in the top quartile rather than in the bottom one. This gap

is even bigger with respect to direct incentives to the answerer. When answerers have high virtual rewards at stake, their increase in quality is five times larger than when they have no stakes.⁴²

9.1. Other Efficiency-Related Trade-Offs

Although the analysis in this paper does not allow us to derive the implications on dimensions other than the contributions' quality, the platform faces additional efficiency trade-offs, namely, on the number of contributions.

First, if users are time-constrained, they may contribute to only one of the sites. By introducing multiple languages, the platform can experience a reduction in the number of contributions in English. Indeed, in the baseline sample, 22% of answerers who were active in English before treatment stopped contributing in English when their native language became available. For similar reasons, other users may maintain participation on multiple sites but publish fewer answers on each site compared with their contribution in English if only the English site were available. The dispersion of content that would be in English or other languages may increase search costs for information seekers or even exclude them from some information. Bao et al. (2012) find evidence of such dispersion on Wikipedia.

Second, the multiplicity of sites may lead to the provision of the same information multiple times. From an efficiency perspective, this is suboptimal, as the concentration of effort on a unique site increases the potential for information aggregation.

Third, introducing the availability of multiple languages may affect the overall community size. On the one hand, the availability of languages other than English may allow users who do not know English to join the platform. This is particularly relevant in Q&A platforms where free riding and undersupply of answers may hinder the success of the site. On the other hand, it may fragment the community based on the language used, creating smaller subcommunities. The reduction in community size may directly impact the number of contributions and cause spillovers from the contributing decisions of the remaining users, as Zhang and Zhu (2011) find for Wikipedia.

Finally, the availability of multiple languages may allow the platform to attract users from competing platforms that use those languages (Jeon et al. 2021).

10. Conclusion

Knowledge platforms like Stack Overflow aim to create a valuable source of information by aggregating knowledge across the internet. Compared with traditional resources like paper encyclopaedias, these platforms face the challenge of ensuring that crowdsourced information is reliable and of high quality.

This paper studied the quality trade-offs knowledge platforms face when making their websites available in one or multiple languages. The results show that the benefits of allowing contributions in multiple languages in addition to English are substantial. Users native to those languages experience a reduction in their cost of communication, leading to higher-quality, more effective contributions. At the same time, these users do not reduce the quality of their contributions in English. However, as language costs act as barriers to participation, the availability of more languages also induces contributions from users who, on average, have lower expertise.

Although this analysis is specific to the context of Stack Overflow, the results may apply to different environments. Extensive literature has addressed communication costs as a major constraint to efficient economic activities; however, to my knowledge, it has not yet quantified the problem. This is relevant to a variety of decision makers. To give a few examples, when firms need to form teams of employees of different nationalities, they need to assess the advantages of pairing coworkers of the same nationality (Lyons 2017, Corritore et al. 2020). In defining the hierarchical structure of the company, managers need to evaluate the advantages of hiring employees who can act as *translators* and allow synergies between teams that use different specialised languages (Crémer et al. 2007). Finally, national states may want to understand the benefits of setting a homogeneous language before limiting individual freedom and the cultural traits of minorities.

Finally, the results provide insights on how the introduction of external technologies, such as live translators and search engines that allow for searches across languages, may impact information quality on the internet (Brynjolfsson et al. 2019). Such technologies, if sufficiently effective, would reduce language barriers. This may benefit knowledge platforms, as they would not have to implement new languages in their design. However, it prevents them from using language barriers as a tool to screen participants.

Future research should be devoted to understanding the external validity of the results in different environments.

Acknowledgments

The author is extremely grateful to three anonymous referees, the associate editor, and the editor for comments that substantially improved this paper. For helpful discussions and feedback, he thanks Patrick Bennett, Milo Bianchi, Gordon Burtch, Daniel L. Chen, Jacques Crémer, Daniel Ershov, April Franco, Guido Friebe, Astrid Hopfensitz, Jakub Lonsky, Nicolas de Roos, Marta Troya-Martinez, Shalini Mitra, and Bobby Zhou. For useful comments, he thanks participants at the University of Liverpool Management

School internal seminar, Digital Economy Network workshop, Munich Summer Institute, Society for Institutional and Organizational Economics 2022, Platform Strategy Research Symposium, and Econometric Society European Meeting—Summer 2022.

Endnotes

¹ The economic theory literature has investigated both constraints but generally focuses on one or the other. On one side, the literature has looked at incentives and strategic information transmission, either with costless communication, that is, the *cheap talk* literature (Crawford and Sobel 1982, Austen-Smith and Banks 2000, Asher and Lascarides 2013, Sobel 2013), or with strategic choice of effort (Dewatripont and Tirole 2005), as in the signalling literature (Spence 1973, Gambetta 2011). On the other side, the team theory literature (Marschak and Radner 1972) has focused on environments where incentives are perfect but exogenous constraints affect the ability to communicate, for example, bounded cognitive abilities or costs in information processing (Arrow 1974, Bolton and Dewatripont 1994, Crémer et al. 2007, Blume and Board 2013, Blume 2018, Dilmé 2018). In this paper, I combine these two strands and examine the interaction between incentives and exogenous costs.

² Stack Overflow is probably the primary source of help for computer programmers. Solutions provided in the platform influence programmers' code worldwide, so mistakes or inaccurate information can have a significant impact. For instance, the most copied code snippet from Stack Overflow contained an error: <https://programming.guide/worlds-most-copied-so-snippet.html>. Although advanced generative artificial intelligence (AI) tools are potentially taking over as the main source of programmers' support, Stack Overflow content is most likely fed into AI algorithms.

³ The cost of language represents the degree of difficulty in using a particular language for communication. In the context of this paper, it mostly corresponds to the inverse of the degree of proficiency in that language. I use this terminology throughout the paper to be consistent with its role as a cost in the theoretical problem.

⁴ Note that the measure is not based on the length of the code. Generally, a shorter code is more efficient, so code length is not a good proxy for answer quality.

⁵ In a nutshell, Borusyak et al. (2024) first estimate the treatment effect for every treated observation and then obtain the average treatment effect on the treated by averaging across them. This approach solves econometric issues identified in the literature (de Chaisemartin and D'Haultfoeuille 2020, Callaway and Sant'Anna 2021, Sun and Abraham 2021, Borusyak et al. 2024).

⁶ Note that this analysis does not consider other possible implications such as trade-offs on the number of contributions and the community size. See Section 9 for a more extensive discussion.

⁷ Yahoo! Answers was shut down in April 2021.

⁸ Nevertheless, the platform has implemented several incentive systems, including virtual rewards.

⁹ See <https://stackoverflow.com/company>.

¹⁰ URLs of the sites are as follows: English: <https://stackoverflow.com/>; Russian: <https://ru.stackoverflow.com/>; Japanese: <https://ja.stackoverflow.com/>; Portuguese: <https://pt.stackoverflow.com/>; and Spanish: <https://es.stackoverflow.com/>.

¹¹ See <https://stackoverflow.blog/2014/02/13/cant-we-all-be-reasonable-and-speak-english/>.

¹² The Section A.1 of the Online Appendix provides more details on the introduction of the websites.

¹³ Users joining when their native language was already available were not directly affected by the introduction of the new sites. I use the users' account registration date to identify when they joined.

¹⁴ This definition of quality is compatible with the one used by Dessein and Santos (2006, section V.A) and Dessein et al. (2016), where communication quality is defined as the probability of a successful communication and is endogenous even though, in their setting, it is exogenous from the communicating parties. By contrast, Dewatripont and Tirole (2005) refer to this probability as the *communication technology*.

¹⁵ An alternative model with similar assumptions is the one presented by Dewatripont and Tirole (2005).

¹⁶ In this context, I use the term *strategy* as used in game theory, where it refers to a function that maps possible values of contingent states (e.g., question quality, topic, language) to a level of answer quality. The assumption of committing to a strategy before observing the information implies that Alice forms her communication strategy while taking expectations over the information that Bob may need. This assumption is realistic if we believe that Alice adopts a long-term strategy that she maintains every time she answers a question. Note that the strategy may still depend on question-specific features and Alice's choice will be different for every answer she writes. The alternative assumption would be that Alice forms her strategy only after observing the information the questioner needs. The latter assumption substantially reduces the tractability of the model, and I do not have specific reasons to believe that this assumption is more accurate than the former.

¹⁷ The precision of the prior can be interpreted as how diverse the possible solutions can be. For instance, because Stack Overflow hosts only questions about programming languages and software, s is a bit higher than what it would be in a platform that hosts any type of question as the range of possible solutions is more limited.

¹⁸ The use of quadratic cost functions, as well as defining the loss function on quadratic distances, is standard in the organisational economics literature, for example, Calvo-Armengol et al. (2015), Dessein and Santos (2006), and Dessein et al. (2016).

¹⁹ Note that, depending on which factors are relevant, Alice's γ may be question specific or fixed across the questions that Alice answers.

²⁰ Details on the steps are provided in Section B of the Online Appendix.

²¹ Supportive evidence includes the fact that 99.8% of users who contribute in both English and other languages contribute in only one non-English language.

²² Note that a natural extension of point 1 is that the effect is larger for users with a lower proficiency in English, as they experience a larger drop in the cost of language when they use their native language. Section G.1 in the Online Appendix studies heterogeneity of the effect across different possible proxies for the cost of using English.

²³ See Section B.4 in the Online Appendix for more details.

²⁴ Figure EC.2 in the Online Appendix shows an example of an answer with two code snippets.

²⁵ A measure based on likes is less preferable, as likes tend to be more sensitive to timings, order of publication of the answers in a question thread, and community size. Nevertheless, accepted answers and answers with more pieces of code have significantly more likes, as shown in Section D.2 in the Online Appendix.

²⁶ The platform advises the inclusion of code for the development of good questions: <https://stackoverflow.com/help/how-to-ask>.

²⁷ There are some cases in which the points are allocated even if the questioner does not select a recipient. In addition, other users who did not write the question can also set up bounties. More details are available at <https://stackoverflow.com/help/privileges/set-bounties>.

²⁸ Note that, in general, other possible explanations do not imply the same prediction, that is, that new joiners contribute lower-quality

content compared with old joiners. For instance, it is possible that, as discussed in Section B.5 in the Online Appendix, topics are systematically different across websites, or users are significantly more empathetic when participating in communities that use their native language. In the former case, new joiners may not have participated in English because of insufficient expertise even though they may have the same expertise as old joiners when using their native language. Similarly, in the latter case, new joiners may not have participated in English because they did not care enough about the questioners' utility in the English community, yet be as empathetic as the old joiners in their native language community. Either way, these explanations imply that new joiners do not write lower-quality content compared with old joiners. This contradicts the theoretical prediction, and it remains an empirical question whether that is true.

²⁹ More precisely, if a user published in both English and, say, Spanish, the user belongs to the treatment group if they published at least (1) one answer in English before the Spanish site became available and (2) one answer in Spanish.

³⁰ The choice of defining the treatment in this way responds to the need to remain close to standard environments, where all observations of treatment group users are considered treated past the treatment date. An alternative strategy would define an answer as treated only when it is written in the native language and exclude English posttreatment answers from the sample. Regression results based on this alternative sample are reported in Section E.1 in the Online Appendix.

³¹ The data are not left-censored, as they have been available since the sites' launch. Nevertheless, some sites had a *proposal* stage for which data are unavailable. For more details, see Section A.1 in the Online Appendix.

³² I discuss the estimation method using *numCodes* as the dependent variable. However, I execute the same estimation also using *Is Best Answer* as the dependent variable.

³³ As discussed in several papers (de Chaisemartin and D'Haultfoeulle 2020, Callaway and Sant'Anna 2021, Goodman-Bacon 2021, Sun and Abraham 2021, Borusyak et al. 2024), the TWFE estimation procedure estimates the treatment effect as a weighted average of all possible treatment effects for each user $\times$ period cell. The weights sum to one but may be negative. If the treatment effects are not homogeneous, asymmetric weighting may cause undesired outcomes.

³⁴ The literature has proposed other solutions. For example, de Chaisemartin and D'Haultfoeulle (2020) and Callaway and Sant'Anna (2021) suggest alternatives that rely only on the data just before and after the treatment of each cohort (i.e., the set of individuals treated at the same time). In the context of this paper, these solutions are less preferable because I observe an unbalanced panel: not all users participate every week. The selection of data may cause the creation of biased comparison groups.

³⁵ This estimation procedure does not support having time periods with treated observations but no untreated observations in the data. This is a situation that could occur at the end of the (unbalanced) panel.

³⁶ Imposing homogeneous weights on the weighted average is not optimal if the objective is to estimate an average effect at the user level. Indeed, this overweighs users that contribute more content. Section E.4 in the Online Appendix provides alternative estimates of the ATT by first obtaining the average effect for each user and then taking the average effect across users, as suggested by Baker et al. (2022).

³⁷ Section C.1 in the Online Appendix provides more details on these variables and the rationale for their inclusion.

³⁸ The mechanism is that questions without any code may not require code in the answer. This fact is discussed in relation to the

answers in Section D.2 in the Online Appendix. It creates a discontinuity on the number of pieces of code in the answers between questions that have and do not have code, which is not addressed by the linearity of the control variable. Section E.5 in the Online Appendix provides additional supporting evidence for this mechanism and the fact that the treatment effect increases with the question's quality. First, it reports the estimates of a model specification that includes question quality as a control variable but excludes questions with no code snippets from the sample. It shows that, in this case, estimates of the treatment effect increase with higher levels of question quality. Second, it reports the estimates obtained including question quality as a control variable and using a dummy equal to one if the questioner accepted the answer as the dependent variable. In this case, the treatment effect increases with the question's quality.

³⁹ The exclusion of questions with only one answer implies the exclusion of about 66% of non-English questions and a 44% reduction of the sample of answers, as shown in Table EC.5 in the Online Appendix. The same table shows that the distribution of the number of answers per question is left skewed. Nevertheless, a significant sample of about 90,000 answers addressed questions that received three or more answers.

⁴⁰ Detailed regression estimates and robustness exercises with alternative quality measures are available in Section F.1 in the Online Appendix.

⁴¹ For further discussions of the trade-offs between efficiency and ethics concerning homogenisation versus diversity of languages, see Ginsburgh and Weber (2011), Blouin and Dyer (2023), and Blanc and Kubo (2024).

⁴² An additional cost on the platform quality trade-off may arise from negative externalities on the English site. Section G.2 in the Online Appendix shows that is not the case and, if anything, there are positive spillovers.

References

- Arrow KJ (1974) *The Limits of Organization* (W. W. Norton & Company, New York).
- Asher N, Lascarides A (2013) Strategic conversation. *Semantics Pragmatics* 6(2):1–62.
- Austen-Smith D, Banks JS (2000) Cheap talk and burned money. *J. Econom. Theory* 91(1):1–16.
- Baker AC, Larcker DF, Wang CCY (2022) How much should we trust staggered difference-in-differences estimates? *J. Financial Econom.* 144(2):370–395.
- Bao P, Hecht B, Carton S, Quaderi M, Horn M, Gergle D (2012) Omnipedia: Bridging the Wikipedia language gap. *Proc. SIG-CHI Conf. Human Factors Comput. Systems* (Association for Computing Machinery, New York), 1075–1084.
- Battiston D, Blanes I, Vidal J, Kirchmaier T (2021) Face-to-face communication in organizations. *Rev. Econom. Stud.* 88(2):574–609.
- Blanc G, Kubo M (2024) French. Working paper, University of Manchester, Manchester, UK.
- Blouin A, Dyer J (2023) How cultures converge: An empirical investigation of linguistic exchange. Working paper, University of Toronto, Toronto, CA.
- Blume A (2018) Failure of common knowledge of language in common-interest communication games. *Games Econom. Behav.* 109:132–155.
- Blume A, Board O (2013) Language barriers. *Econometrica* 81(2): 781–812.
- Blume A, DeJong DV, Kim YG, Sprinkle GB (2001) Evolution of communication with partial common interest. *Games Econom. Behav.* 37(1):79–120.
- Bolton P, Dewatripont M (1994) The firm as a communication network. *Quart. J. Econom.* 109(4):809–839.

- Borusyak K, Jaravel X, Spiess J (2024) Revisiting event-study designs: robust and efficient estimation. *Rev. Econom. Stud.* rdae007.
- Boudreau KJ, Brady T, Ganguli I, Gaule P, Guinan E, Hollenberg A, Lakhani KR (2017) A field experiment on search costs and the formation of scientific collaborations. *Rev. Econom. Statist.* 99(4):565–576.
- Brynjolfsson E, Hui X, Liu M (2019) Does machine translation affect international trade? Evidence from a large digital platform. *Management Sci.* 65(12):5449–5460.
- Callaway B, Sant'Anna PHC (2021) Difference-in-differences with multiple time periods. *J. Econometrics* 225(2):200–230.
- Calvó-Armengol A, de Martí J, Prat A (2015) Communication and influence. *Theoret. Econom.* 10(2):649–690.
- Chen S, Gelyukens R, Choi CJ (2006) The importance of language in global teams: A linguistic perspective. *Management Internat. Rev.* 46(6):679–696.
- Corritore M, Goldberg A, Srivastava SB (2020) Duality in diversity: How intrapersonal and interpersonal cultural heterogeneity relate to firm performance. *Admin. Sci. Quart.* 65(2):359–394.
- Crawford VP, Sobel J (1982) Strategic information transmission. *Econometrica* 50(6):1431–1451.
- Crémer J, Garicano L, Prat A (2007) Language and the theory of the firm. *Quart. J. Econom.* 122(1):373–407.
- de Chaisemartin C, D'Haultfoeuille X (2020) Two-way fixed effects estimators with heterogeneous treatment effects. *Amer. Econom. Rev.* 110(9):2964–2996.
- Dessein W, Santos T (2006) Adaptive organizations. *J. Political Econom.* 114(5):956–995.
- Dessein W, Galeotti A, Santos T (2016) Rational inattention and organizational focus. *Amer. Econom. Rev.* 106(6):1522–1536.
- Dewatripont M, Tirole J (2005) Modes of communication. *J. Political Econom.* 113(6):1217–1238.
- Dilmé F (2018) Optimal languages. Working paper, University of Bonn, Bonn, Germany.
- Gambetta D (2011) *Codes of the Underworld: How Criminals Communicate* (Princeton University Press, Princeton, NJ).
- Ginsburgh V, Weber S (2011) *How Many Languages Do We Need?: The Economics of Linguistic Diversity* (Princeton University Press, Princeton, NJ).
- Goldfarb A, Tucker C (2019) Digital economics. *J. Econom. Lit.* 57(1): 3–43.
- Goodman-Bacon A (2021) Difference-in-differences with variation in treatment timing. *J. Econometrics* 225(2):254–277.
- Guillouët L, Khandelwal AK, Macchiavello R, Teachout M (2024) Language barriers in multinationals and knowledge transfers. *Rev. Econom. Statist.* 1–56.
- Jeon DS, Jullien B, Klimenko M (2021) Language, internet and platform competition. *J. Internat. Econom.* 131:103439.
- Lafky J, Wilson AJ (2020) Experimenting with incentives for information transmission: Quantity vs. quality. *J. Econom. Behav. Organ.* 169:314–331.
- Lohmann J (2011) Do language barriers affect trade? *Econom. Lett.* 110(2):159–162.
- Lyons E (2017) Team production in international labor markets: Experimental evidence from the field. *Amer. Econom. J. Appl. Econom.* 9(3):70–104.
- Marschak J, Radner R (1972) *Economic Theory of Teams* (Yale University Press, New Haven, CT).
- McManus WS (1985) Labor market costs of language disparity: An interpretation of Hispanic earnings differences. *Amer. Econom. Rev.* 75(4):818–827.
- Melitz J (2008) Language and foreign trade. *Eur. Econom. Rev.* 52(4): 667–699.
- Sandvik JJ, Saouma RE, Seegert NT, Stanton CT (2020) Workplace knowledge flows. *Quart. J. Econom.* 135(3):1635–1680.
- Sobel J (2013) Giving and receiving advice. Acemoglu D, Arellano M, Dekel E, eds. *Advances in Economics and Econometrics*, Econometric Society Monographs (Cambridge University Press, Cambridge, UK), 305–341.
- Spence M (1973) Job market signaling. *Quart. J. Econom.* 87(3):355–374.
- Spolsky J (2008) Stack overflow launches. *Joel on Software* (blog) (September 15), <https://www.joelonsoftware.com/2008/09/15/stack-overflow-launches/>.
- Sun L, Abraham S (2021) Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *J. Econometrics* 225(2):175–199.
- Tainer E (1988) English language proficiency and the determination of earnings among foreign-born men. *J. Human Resources* 23(1): 108–122.
- Tenzer H, Pudelko M, Harzing AW (2014) The impact of language barriers on trust formation in multinational teams. *J. Internat. Bus. Stud.* 45(5):508–535.
- Zhang XM, Zhu F (2011) Group size and incentives to contribute: A natural experiment at Chinese Wikipedia. *Amer. Econom. Rev.* 101(4):1601–1615.